

Consider the following polyhedron represented in standard form:

$$P = \left\{ x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mid Ax = b, x \geq 0 \right\},$$

with $A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Find the basic directions for the following feasible basic solutions, and mention if they are feasible:

1. Basic solution with x_2 and x_4 in the basis ($j_1 = 2, j_2 = 4$):

$$B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}.$$

2. Basic solution with x_1 and x_4 in the basis ($j_1 = 1, j_2 = 4$):

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

3. Basic solution with x_1 and x_2 in the basis ($j_1 = 1, j_2 = 2$):

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}; B^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$